

Q1.

1. Select different types of Elastic load balancers available in AWS. – 4M

Elastic Load Balancing (ELB) in AWS is a service used to automatically distribute incoming application traffic across multiple targets such as EC2 instances, containers, IP addresses, and Lambda functions

1. Application Load Balancer (ALB)

Application Load Balancer operates at **Layer 7 (Application Layer)** of the OSI model. It is designed for HTTP and HTTPS traffic.

Features:

- Performs **content-based routing**, such as directing requests based on URL path, hostname, headers, or query strings.
- Supports **microservices and containerized applications**.
- Provides **SSL/TLS termination**.
- Supports **WebSocket and HTTP/2**.
- Can route traffic to multiple target groups.

Use Case:

Used in web applications where requests like /images and /videos need to go to different servers.

2. Network Load Balancer (NLB)

NLB operates at **Layer 4 (Transport Layer)** and handles TCP, UDP, and TLS traffic.

Features:

- Very high performance and low latency.
- Can handle millions of requests per second.
- Supports **static IP addresses**.
- Suitable for sudden traffic spikes.
- Preserves source IP address.

Use Case:

Used for gaming apps, IoT, or high-performance applications.

3. Gateway Load Balancer (GWLB)

Operates at **Layer 3 and Layer 4** and is designed for deploying security appliances.

Features:

- Used with virtual firewalls and intrusion detection systems.
- Provides transparent scaling of security devices.
- Combines load balancing with packet routing.

Use Case:

Useful in network security architectures.

4. Classic Load Balancer (CLB)

Legacy load balancer supporting both Layer 4 and Layer 7.

Features:

- Basic load balancing.
- Supports EC2 Classic.
- Suitable for older generation applications.

Use Case:

Used mainly in legacy systems, though AWS recommends ALB or NLB.

Q1.

2. Purpose of CloudWatch Logs in tracking server performance. – 4M

Amazon CloudWatch Logs is an AWS monitoring service used to collect, store, analyze, and monitor log files generated by servers, applications, and AWS services

Purpose of CloudWatch Logs

1. Centralized Log Collection

CloudWatch Logs gathers logs from:

- EC2 instances
- Application servers
- Lambda functions
- Containers
- System and application logs

2. Performance Monitoring

Logs help track important performance indicators such as:

- CPU usage patterns
- Memory-related warnings
- Disk utilization issues
- Network traffic anomalies
- Slow application responses

3. Troubleshooting and Error Detection

CloudWatch Logs captures:

- Error messages
- Crash reports
- Failed login attempts
- Server exceptions
- Application failures

4. Log Analysis with Metrics and Alarms

CloudWatch can convert log data into **custom metrics**.

Example:

- Count number of failed requests.
- Trigger alarms if error rate crosses threshold.
- Send notifications using Amazon SNS.

5. Real-Time Monitoring

CloudWatch Logs provides near real-time log streaming.

Benefits:

- Detect unusual behavior instantly.
- Monitor live server events.
- Respond quickly to incidents.

6. Security and Auditing

It helps monitor suspicious activities:

- Unauthorized access attempts
- Security events
- Compliance auditing

Q1.

3. Virtualization in the Context of Amazon EC2 Instances. – 4M

Virtualization is a technology that allows multiple virtual machines to run on a single physical server by sharing its hardware resources. In the context of **Amazon EC2**, virtualization is the foundation that enables users to launch scalable virtual servers called **EC2 instances**.

Meaning of Virtualization in EC2

Instead of dedicating one physical server to one user, AWS uses virtualization to divide physical hardware into multiple isolated virtual servers.

Each EC2 instance gets:

- Virtual CPU (vCPU)
- Virtual memory (RAM)
- Virtual storage
- Virtual networking

Benefit of virtualization in EC2

1. Resource Sharing

Multiple EC2 instances share one physical server efficiently, reducing cost.

2. Scalability

Users can launch or terminate instances on demand based on workload.

3. Isolation

Each instance runs independently, improving security and fault isolation.

4. Flexibility

Different operating systems and configurations can run on the same hardware.

5. Cost Efficiency

Users pay only for resources used, avoiding physical hardware expenses.

Example

A company can run:

- Web server on one EC2 instance
- Database server on another
- Application server on a third

All may run on shared AWS physical infrastructure through virtualization.

Q2.

1. Organize how CloudWatch Essentials and Serverless Architectures contribute to IaaS principles and operational efficiency. -8M

Amazon CloudWatch Essentials and **Serverless Architectures** play a major role in supporting **Infrastructure as a Service (IaaS)** principles while improving operational efficiency in cloud environments. Together they provide automation, monitoring, scalability, reliability, and cost optimization.

Contribution of CloudWatch Essentials to IaaS

Amazon CloudWatch is a monitoring and observability service used to track infrastructure resources and application performance.

1. Monitoring and Visibility

CloudWatch monitors:

- EC2 performance
- CPU and memory usage
- Disk activity
- Network traffic
- Application logs and events

This visibility helps administrators manage infrastructure efficiently.

2. Performance Optimization

CloudWatch metrics help identify:

- Resource bottlenecks
- High utilization
- Unused resources
- Performance degradation

This supports efficient resource usage, which is a core IaaS principle.

3. Automation through Alarms

CloudWatch alarms can trigger:

- Auto Scaling actions
- Notifications using SNS

- Automated responses using Lambda

This reduces manual intervention and improves operational efficiency.

4. Fault Detection and Reliability

CloudWatch detects failures early through logs and metrics, improving system availability and fault tolerance.

Contribution of Serverless Architectures

Serverless architecture allows developers to run applications without managing servers.

1. Reduced Infrastructure Management

Cloud provider manages:

- Servers
- Scaling
- Patching
- Maintenance

This aligns with IaaS goals of simplifying infrastructure handling.

2. Automatic Scalability

Serverless services like AWS Lambda scale automatically according to demand, improving performance and resource efficiency.

3. Cost Efficiency

Users pay only for actual execution or usage, reducing infrastructure costs.

4. Faster Deployment

Applications can be deployed quickly without provisioning servers, improving agility.

Q2.

2. Simplify “Lambda Essentials” and explain how they relate to serverless architecture. – 4M

AWS Lambda Essentials refers to the core concepts and features of AWS Lambda, a serverless computing service that allows code execution without managing servers. It is a key component of **serverless architecture**, where infrastructure management is handled by the cloud provider.

Lambda Essentials

1. Event-Driven Execution

Lambda runs when triggered by events such as:

- File upload in S3
- API requests through API Gateway
- Database updates
- CloudWatch events
- SNS notifications

2. No Server Management

Users do not manage:

- Servers
- Operating systems
- Patching
- Capacity planning

3. Automatic Scaling

- Lambda automatically scales according to the number of incoming requests.

4. Pay-Per-Use Model

Charges are based on:

- Number of requests
- Execution duration

Relation to Serverless Architecture

Lambda is closely related to serverless architecture because it is one of its core building blocks.

Relation Points

- **Implements serverless model:** No need to manage servers.
- **Supports event-driven architecture:** A fundamental principle of serverless.
- **Provides automatic scaling:** Core feature of serverless systems.
- **Enables microservices:** Applications can be divided into independent functions.
- **Reduces operational overhead:** Infrastructure management is handled by AWS.
- **Improves cost efficiency:** Pay-per-use aligns with serverless economics.

Q2.

1. Analyse the role of Route 53 Essentials and VPC Networking in Domain Name Resolution and Service Delivery. – 8M

Role of Route 53 Essentials:

Route 53 is AWS's **Domain Name System (DNS)** service.

1. Domain Name Resolution

It translates domain names like:

www.example.com

into IP addresses users can reach.

2. Traffic Routing

Route 53 supports advanced routing:

- Simple routing
- Weighted routing
- Latency-based routing
- Geolocation routing
- Failover routing

3. High Availability

If one server fails, failover routing redirects users to healthy resources.

4. Health Checks

Route 53 monitors endpoint health and routes traffic only to healthy servers.

Role of VPC Networking

A **Virtual Private Cloud (VPC)** provides isolated networking for AWS resources.

1. Secure Communication

VPC enables secure communication between:

- EC2 instances
- Databases
- Applications

2. Subnets and Routing

Public and private subnets organize resources and control traffic flow.

3. Security Control

Security Groups and Network ACLs protect services from unauthorized access.

4. Internal Name Resolution

VPC supports private DNS for communication between internal resources.

Q2.

2. Conclude the function of an “Internet Gateway” in a custom VPC and explain briefly. – 4M

An **Internet Gateway (IGW)** is a VPC component that allows communication between resources in a VPC and the public internet. It is essential for internet connectivity in a custom VPC.

Functions of Internet Gateway

1. Enables Internet Access

It allows EC2 instances in public subnets to:

- Access the internet
- Download updates
- Serve websites to users

2. Supports Incoming and Outgoing Traffic

It handles:

- Inbound traffic from internet users
- Outbound traffic from VPC resources

3. Works with Route Tables

To function, route tables need a route such as:

0.0.0.0/0 → Internet Gateway

4. Provides One-to-One NAT for Public IPs

Instances with public or Elastic IPs can communicate externally through the Internet Gateway.

5. Supports Public Subnets

A subnet becomes public only when:

- It has route to IGW
- Instances have public IPs

Example

A web server in a public subnet uses:

- Security Group for access control
- Route table for internet route
- Internet Gateway for connectivity

Q3.

1. Determine an architecture for an “Automated Security Monitoring” system that utilizes Flow Logs, CloudWatch and Lambda. – 8M

An **Automated Security Monitoring System** in AWS can be designed by integrating **VPC Flow Logs, Amazon CloudWatch, and AWS Lambda** to detect threats, monitor suspicious activities, and respond automatically. This architecture improves security, automation, and incident response.

Architecture Components

1. VPC Flow Logs (Traffic Monitoring Layer)

VPC Flow Logs capture information about network traffic entering and leaving network interfaces in a VPC.

They monitor:

- Source and destination IPs
- Ports and protocols
- Allowed or rejected traffic
- Suspicious access attempts
- Unusual traffic patterns

Example:

Repeated failed connections from unknown IPs may indicate an attack attempt.

2. CloudWatch (Monitoring and Alert Layer)

Flow Log data is sent to **Amazon CloudWatch Logs** for monitoring and analysis.

CloudWatch performs:

- Log collection and storage
- Metric creation from log patterns
- Security event detection
- Alarm generation

CloudWatch can detect:

- Port scanning
- Unauthorized access attempts
- High traffic spikes

- Abnormal network behavior

Example:

If rejected traffic exceeds a threshold, CloudWatch alarm triggers.

3. AWS Lambda (Automated Response Layer)

When CloudWatch alarms detect suspicious events, they trigger **AWS Lambda** automatically.

Lambda can perform actions like:

- Block malicious IPs using Security Groups or NACLs
- Send alerts using SNS
- Trigger remediation workflows
- Stop compromised EC2 instances
- Generate incident reports

Example:

If brute-force login attempts are detected, Lambda can automatically block the attacker IP.

Architecture Workflow

Architecture can be organized as:

Network Traffic → VPC Flow Logs → CloudWatch Monitoring & Alarms → Lambda Automated Response

Steps:

1. Flow Logs capture traffic data.
2. CloudWatch analyzes logs.
3. Alarm detects suspicious pattern.
4. Lambda executes automatic security action.

Benefits

- Real-time threat monitoring
- Automated incident response
- Reduced manual effort
- Improved security and operational efficiency

Q3.

2. Examine “Route 53 DNS Failover” and its goal and explain briefly. – 4M

Route 53 DNS Failover is a feature of **Amazon Route 53** that automatically redirects traffic from an unhealthy resource to a healthy backup resource when failure occurs. Its main goal is to ensure **high availability and continuous service delivery**.

How It Works

Step 1: Configure Primary and Secondary Resources

Example:

- Primary web server in one region
- Backup server in another region

Step 2: Health Checks

Route 53 checks:

- Server availability
- Response time
- Application health

Step 3: Automatic Failover

If primary server fails:

- Health check reports failure
- DNS records switch traffic to backup server
- Users continue accessing service

Goal of Route 53 DNS Failover

1. High Availability

Keeps applications available even during server failures.

2. Fault Tolerance

Provides backup routing when resources become unavailable.

3. Disaster Recovery

Supports recovery during outages or regional failures.

4. Reduced Downtime

Minimizes service interruption.

5. Reliable Service Delivery

Ensures users can access applications continuously.

Example

If an e-commerce website's primary server goes down, Route 53 automatically redirects customers to a backup server, preventing downtime.

Q3.

1. Not available

Q3.

2. Inspect the appropriate Tool for Analyzing “CPU and Memory Usage” Over Time in AWS. – 4M

The most appropriate AWS tool for analyzing **CPU and Memory Usage over time** is **Amazon CloudWatch**. It is a monitoring and observability service used to collect, track, visualize, and analyze performance metrics of AWS resources in real time and historically.

Why Amazon CloudWatch is Appropriate

1. Performance Metrics Monitoring

CloudWatch monitors important system metrics such as:

- CPU Utilization
- Memory Usage
- Disk Activity
- Network Traffic

2. Analysis Over Time

CloudWatch stores historical metric data, allowing users to analyze trends over time.

It helps observe:

- Daily or weekly CPU patterns
- Memory consumption growth
- Resource bottlenecks
- Performance degradation trends

3. CloudWatch Metrics and Graphs

CloudWatch provides:

- Real-time monitoring graphs
- Custom dashboards
- Metric visualization
- Historical performance reports

4. Alarms and Alerts

CloudWatch allows creation of alarms.

Example:

- Alert if CPU exceeds 80%
- Notify if memory usage crosses threshold

5. Auto Scaling Support

Based on CPU metrics, CloudWatch can trigger **Auto Scaling** to add or remove resources automatically.

Example

If an EC2 server shows increasing CPU usage during peak hours, CloudWatch graphs can reveal the trend and help decide whether scaling is needed.

Q4.

1. Categorize the significance of S3 components (buckets and objects) in a scalable architecture. – 8M

Amazon S3 (Simple Storage Service) is a highly scalable object storage service in AWS. Its two main components, **Buckets** and **Objects**, play a vital role in building scalable, reliable, and efficient cloud architectures.

1. Buckets and Their Significance

A **Bucket** is a container used to store objects in S3. It acts as the top-level organizational unit.

Importance of Buckets in Scalable Architecture

a) Data Organization

Buckets help organize data by:

- Application type
- Project
- Environment (Dev, Test, Production)
- Department or user group

b) Scalability

Buckets support virtually unlimited storage, allowing applications to scale without worrying about storage capacity.

c) Access Control and Security

Bucket policies and permissions control access to data securely, important in scalable distributed systems.

d) Availability and Durability

Buckets benefit from S3's high durability and availability, supporting reliable large-scale architectures.

2. Objects and Their Significance

An **Object** is the actual data stored in a bucket, such as files, images, videos, backups, or logs.

Importance of Objects in Scalability

a) Massive Data Storage

Objects allow storage of huge amounts of structured or unstructured data, supporting big scalable applications.

b) Independent Access

Each object has a unique key and can be retrieved independently, improving efficiency and performance.

c) Metadata Support

Objects store metadata that helps manage, classify, and retrieve data effectively.

d) Lifecycle Management

Objects can use lifecycle rules for archival or deletion, optimizing storage cost at scale.

Role in Scalable Architecture

Together Buckets and Objects support:

- Elastic storage growth
- High availability
- Cost optimization
- Content delivery
- Backup and disaster recovery
- Support for big data and cloud applications

Example:

A global application can store millions of user images as objects inside scalable S3 buckets.

Q4.

2. Justify the different CloudWatch Monitoring Tasks and Their Importance in Server Delivery. – 4M

Amazon CloudWatch is an AWS monitoring service used to observe system performance, collect metrics, generate alerts, and improve server operations

CloudWatch Monitoring Tasks

1. Resource Monitoring

CloudWatch monitors server resources such as:

- CPU utilization
- Memory usage
- Disk activity
- Network traffic

2. Log Monitoring

CloudWatch Logs collect and analyze:

- Application logs
- System logs
- Error logs
- Security events

3. Performance Analysis

CloudWatch tracks performance trends over time to detect:

- Bottlenecks
- Overloaded servers
- Underutilized resources

4. Alarms and Alerts

CloudWatch alarms notify administrators when thresholds are crossed.

Example:

- High CPU usage alerts
- Low disk space warnings
- Network anomaly alerts

5. Auto Scaling Support

CloudWatch metrics can trigger Auto Scaling.

Example:

If CPU rises above threshold, additional instances can be launched automatically.

6. Event Monitoring and Automation

CloudWatch Events can trigger automated actions using Lambda for corrective responses.

Importance in Server Delivery

CloudWatch monitoring tasks improve server delivery by:

a) High Availability

Detects failures early and reduces downtime.

b) Performance Optimization

Ensures servers operate efficiently.

c) Reliability

Continuous monitoring improves dependable service delivery.

d) Scalability

Supports demand-based scaling.

e) Faster Troubleshooting

Logs and metrics help solve issues quickly.

Q4.

1. Conclude the steps to host a secure static website on S3 using CLI and bucket policies. – 8M

Amazon S3 can be used to host a **static website** containing HTML, CSS, JavaScript and media files

Steps to Host the Website

1. Create an S3 Bucket

First create a bucket to store website files.

Using AWS CLI:

```
aws s3 mb s3://my-static-website
```

The bucket name should be globally unique.

2. Upload Website Files

Upload files like:

- index.html
- error.html
- CSS files
- Images
- JavaScript files

CLI command:

```
aws s3 cp ./website s3://my-static-website/ --recursive
```

3. Enable Static Website Hosting

Configure the bucket for website hosting:

```
aws s3 website s3://my-static-website \  
--index-document index.html \  
--error-document error.html
```

4. Configure Bucket Policy

Bucket policy grants controlled access to website objects.

Example policy:

```
{
  "Version": "2026-04-28",
  "Statement": [{
    "Effect": "Allow",
    "Principal": "*",
    "Action": "s3:GetObject",
    "Resource": "arn:aws:s3::my-static-website/*"
  }]
}
```

Attach policy using CLI or console.

5. Apply Security Measures

For secure hosting:

- Enable **Block Public Access** carefully
- Use **least privilege policies**
- Enable **bucket encryption**
- Enable **versioning**
- Use **CloudFront + HTTPS** for secure delivery
- Restrict write permissions

6. Test Website Endpoint

Access the generated S3 website endpoint and verify proper loading.

Q4.

2. Identify how Security Groups and Network ACLs Provide Layered VPC Security. – 4M

In AWS VPC, **Security Groups** and **Network ACLs (NACLs)** work together to provide **layered security** by protecting resources at different levels.

1. Security Groups

A **Security Group** acts as a **virtual firewall at the instance level**.

It controls traffic to and from resources like EC2 instances.

Functions of Security Groups

They define:

- Allowed inbound traffic
- Allowed outbound traffic
- Port-level access rules
- Protocol access controls

Example:

- Allow HTTP on port 80
- Allow SSH on port 22 only from admin IP

Characteristics

- Stateful firewall
- If inbound is allowed, return traffic is automatically allowed.
- Works at instance level.

2. Network ACLs (NACLs)

A **Network ACL** acts as a firewall at the **subnet level**.

It controls traffic entering and leaving subnets.

Functions of NACLs

They:

- Allow or deny traffic
- Filter traffic at subnet boundary
- Control inbound and outbound rules

Example:

- Deny suspicious IP ranges
- Allow only specific ports

Characteristics

- Stateless firewall
- Inbound and outbound rules configured separately.
- Supports allow and deny rules.

3. How They Work Together (Layered Security)

First Layer – Network ACL

Traffic first passes through NACL rules at subnet level.

Second Layer – Security Group

Allowed traffic then reaches the EC2 instance where Security Group rules apply.

Example

For a web server:

- NACL allows only HTTP/HTTPS traffic to subnet.
- Security Group allows only port 80/443 to EC2.

Q5.

1. Evaluate the use of Bucket Policies versus IAM User Policies for S3 Access Control. - 6M

1. Bucket Policies

A **Bucket Policy** is a **resource-based policy** attached directly to an S3 bucket.

It defines **who can access the bucket and what actions they can perform.**

Uses of Bucket Policies

Bucket policies are useful for:

- Granting access to specific users or accounts
- Public read access for static websites
- Cross-account access
- Restricting access by IP address
- Enforcing encryption or security rules

Example permissions:

- Allow GetObject
- Deny unencrypted uploads
- Allow access only from certain networks

Advantages

- Applied directly to the bucket
- Good for shared or public access scenarios
- Supports cross-account permissions
- Easier centralized control over bucket resources

2. IAM User Policies

IAM User Policies are **identity-based policies** attached to users, groups, or roles.

They define **what actions a user can perform on S3 resources.**

Uses of IAM Policies

Used for:

- Granting user-specific permissions

- Controlling access based on roles
- Applying least privilege security
- Managing internal user access

Example:

A user may have permission only to upload files but not delete them.

Advantages

- Fine-grained user-level control
- Strong identity management
- Easy to manage employee permissions
- Integrates with roles and authentication

Feature	Bucket Policy	IAM User Policy
Type	Resource-based	Identity-based
Attached To	S3 Bucket	User/Role/Group
Controls	Who can access bucket	What user can do
Cross-account Access	Yes	Limited
Public Access	Yes	No
User-specific Control	Limited	Strong

Q5.

2. Relationship Between AWS Account Management and the Services Layer in AWS Hierarchy. – 6M

1. AWS Account Management

AWS Account Management refers to managing:

- AWS accounts
- Users and permissions
- Billing and costs
- Security settings
- Resource ownership
- Account-level governance

Functions of Account Management

It provides:

- IAM users and roles
- Access control
- Billing management
- Multi-account organization
- Security policies

2. Services Layer in AWS Hierarchy

The Services Layer includes AWS cloud services such as:

- EC2
- S3
- Lambda
- RDS
- VPC
- CloudWatch

3. Relationship Between Them

Account as Access Foundation

AWS services are accessed only through an AWS account.

Without an account, services cannot be provisioned or managed.

Account Controls Service Usage

Account management governs:

- Who can use services
- What services they can access
- Resource permissions
- Security policies applied to services

Example:

IAM policies control access to EC2 or S3.

Billing and Resource Ownership

All services used belong to an AWS account, and usage charges are tracked through that account.

Security Integration

Account-level security protects service usage through:

- IAM
- MFA
- Policies
- Roles

Q5.

1. Construct and contrast S3 standard storage with s3 glacier for organizational backup strategies. – 6M

Amazon S3 Standard and **Amazon S3 Glacier** are AWS storage classes used in backup strategies, but they serve different purposes based on access frequency, cost, and retrieval speed

1. Amazon S3 Standard

S3 Standard is designed for **frequently accessed data** that requires high availability and fast retrieval.

Features

- Millisecond data access
- High availability and durability
- Immediate retrieval
- Suitable for active backups
- Supports regular read/write operations

Use in Backup Strategies

Used for:

- Active backups
- Frequently restored data
- Application backups
- Disaster recovery data needing quick access

Advantages

- Fast access to backup files
- High performance
- Reliable for operational workloads

Limitation

- Higher storage cost compared to archival classes.

2. Amazon S3 Glacier

S3 Glacier is designed for **long-term archival and infrequently accessed backups**.

Features

- Very low storage cost
- Designed for archive data
- Retrieval can take minutes to hours
- High durability
- Suitable for compliance storage

Use in Backup Strategies

Used for:

- Long-term backups
- Compliance records
- Historical archives
- Disaster recovery copies rarely accessed

Advantages

- Cost-effective archival storage
- Ideal for retaining large backup volumes

Limitation

- Slower retrieval compared to S3 Standard.

Feature	S3 Standard	S3 Glacier
Access Frequency	Frequent	Rare
Retrieval Speed	Immediate	Slow
Storage Cost	Higher	Lower
Backup Type	Active backups	Archive backups
Performance	High	Lower
Best Use	Operational data	Long-term retention

Q5.

2. How EC2 Placement Groups Enhance Application Resilience and Performance - 6M

EC2 Placement Groups are a feature in AWS that influence how EC2 instances are physically placed within AWS infrastructure

AWS provides three main types of placement groups.

1. Cluster Placement Group

In a cluster placement group, instances are placed close together within the same Availability Zone.

Purpose

Enhances:

- Low network latency
- High throughput
- Fast instance communication

Use Cases

Used for:

- High performance computing
- Big data workloads
- Distributed databases

Performance Benefit

Because instances are close together, communication speed increases significantly.

2. Spread Placement Group

Spread placement groups place instances on separate underlying hardware.

Purpose

Improves resilience by reducing risk of multiple instance failures.

Benefits

- Fault isolation
- Protection against hardware failure
- High availability

Use Cases

Critical applications where each instance must remain isolated.

Example:

Primary and backup servers.

3. Partition Placement Group

Instances are distributed across logical partitions, each with separate hardware.

Purpose

Provides both scalability and resilience.

Benefits

- Failure isolation by partition
- Supports distributed workloads
- Improves fault tolerance

Use Cases

- Hadoop
- Cassandra
- Large distributed systems

How They Enhance Performance

Placement groups improve:

- Network performance
- Low latency communication
- High throughput workloads
- Efficient distributed processing